

(1001CJA102120066)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)

FULL SYLLABUS

11-04-2021

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)**ANSWER KEY****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	D	A	A	C	B	A	D	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	D	B	B	B	B	A	B	A
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.41	10.00	13.33	4.50	1.73	2.50	1.00	1.50	1.50	4.09 to 4.11

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	B	C	C	A	C	D	A	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	A	B	A	A	A	A	B	A
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	9.00	326.40	3.00	4.00	7.57 OR 7.58	7.00	5.00	9.00	4.00	6.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	C	C	C	D	C	A	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	B	B	A	D	D	C	C	B
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	13.00	4.00	8.00	0.60	2.00	6.00	2.00	3.00	2.50	4.00

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)**PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A)****Sol.** Wave is propagating in -ve x-direction.

$$\therefore \hat{E} \times \hat{B} = \hat{V}$$

$$\hat{K} \times \hat{B} = -\hat{i}$$

$$\text{So } \hat{B} = \hat{j}$$

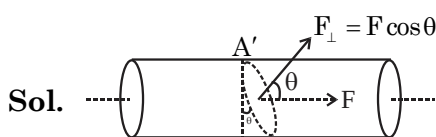
$$B_0 = \frac{E_0}{C} = \frac{1.5 \times 10^6}{3 \times 10^8} = 5 \times 10^{-3} \text{ T}$$

$$B = (B_0 \hat{j}) \sin[\omega t + kx + \phi]$$

Or

$$\vec{B} = (B_0 \hat{j}) \cos[\omega t + kx + \phi]$$

$$\vec{B} = (5 \times 10^{-3} \hat{j}) \cos(\omega t + kx)$$

2. **Ans. (A)**

$$A' \cos \theta = A$$

$$A' = A / \cos \theta$$

$$\text{Tensile stress} = \frac{F_{\perp}}{A'} = \frac{F \cos \theta}{A / \cos \theta} = \frac{F}{A} \cos^2 \theta$$

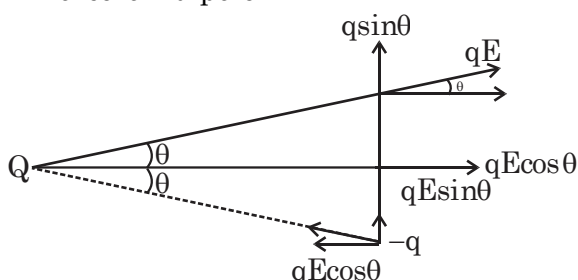
3. **Ans. (D)****Sol.** $\vec{\tau} = \vec{P} \times \vec{E}$ Here $\vec{P} \perp \vec{E}$

$$\tau = PE \sin 90^\circ = PE$$

$$\tau = P \cdot \frac{KQ}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{PQ}{r^2}$$

Sense of rotation $\Rightarrow \vec{P} \times \vec{E} \Rightarrow$ clockwiseDirection of torque $\Rightarrow \otimes$

Force on dipole



$$F_{\text{net}} = 2qE \sin \theta \quad \{\theta \ll\}$$

(upward)

$$= (2qE) \cdot \frac{d/2}{r} \quad \{\sin \theta = \theta\} = \frac{Eqd}{r} = \frac{KQ}{r^2} \cdot \frac{qd}{r}$$

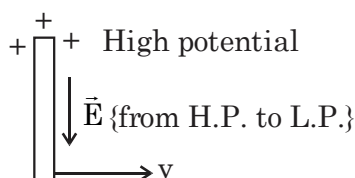
$$F_{\text{net}} = \frac{1}{4\pi\epsilon_0} \frac{QP}{r^3}, \text{ So } \tau = Fr_1$$

$$= \frac{QP}{4\pi\epsilon_0 r^3} \cdot r = \frac{QP}{4\pi\epsilon_0 r^2}$$

4. **Ans. (A)****Sol.** D_2 is reversed bias therefore no current from 10Ω resistance.5. **Ans. (A)**

$$\text{Sol. } F_t = 15 \times 10^9 \times \frac{15}{100} = 225 \times 10^7 \text{ Hz}$$

$$\text{Channels} = \frac{225 \times 10^7}{5 \times 10^3} = 4.5 \times 10^5$$

6. **Ans. (C)****Sol.** In convex lens real image is inverted.7. **Ans. (B)****Sol.**

- at low potential

$$\text{emf induced} = \vec{\ell} \cdot [\vec{v} \times \vec{B}]$$

8. **Ans. (A)****Sol.** $P = P_0 + \rho gh$ as, $h \uparrow, P \uparrow$, so we require more strength at bottom.9. **Ans. (D)**

$$\text{Sol. } t = \frac{xy^2}{z^3}$$

$$\frac{\Delta t}{t} = \left| \frac{\Delta x}{x} \right| + 2 \left| \frac{\Delta y}{y} \right| + 3 \left| \frac{\Delta z}{z} \right| = 1 + 2(3) + 3(2)$$

$$= 1 + 6 + 6 = 13\%$$

10. Ans. (A)

Sol. No current in bulb 4, 5 and 6.

11. Ans. (B)

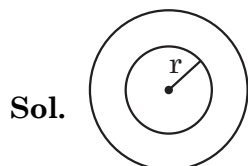
Sol. Apply colm $\{F_{\text{net}} = \text{zero}\}$

$$m_1 v_1 + m_2 v_2 = 0$$

$$0.3(80) + 60(v_2) = 0$$

$$v_2 = \frac{-24}{60} = -0.4 \text{ m/s}$$

12. Ans. (A)



Sol.

$$m = iA$$

$$i = \frac{dQ}{dt} = \frac{\sigma 2\pi r dr}{T}$$

$$m = \frac{\sigma(2\pi r) dr}{T} \cdot \pi r^2$$

$$= \int_0^R \sigma \omega (\pi) r^3 \cdot dr = \frac{\sigma \omega \pi}{4} R^4$$

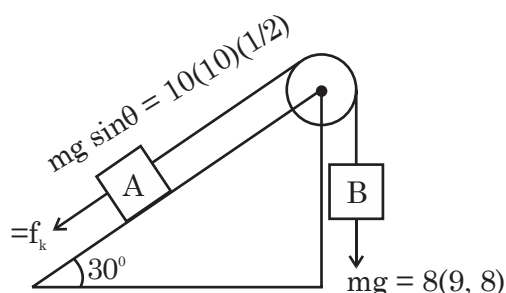
$$|\tau| = |\vec{m} \times \vec{B}| = mB \sin \theta$$

$$= \frac{1}{4} \sigma \omega \pi R^4 B \sin \theta$$

13. Ans. (D)

14. Ans. (B)

Sol. Net pulling force along string



$$= 0.2(10)(9.8) \frac{\sqrt{3}}{2}$$

Net pulling force along string

$$= 8(9.8) - 10(9.8) \cdot \frac{1}{2} - 0.2(10)(9.8) \frac{\sqrt{3}}{2}$$

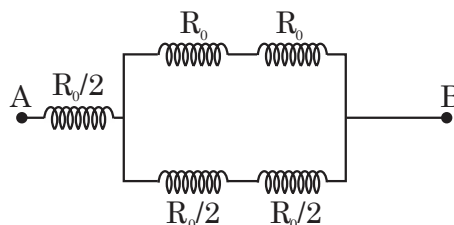
$$a = \frac{F_{\text{net}}}{M_{\text{total}}} = \frac{F_{\text{net}}}{18} = \frac{12.42}{18}$$

$$\approx 0.69 \text{ m/s}^2 \text{ (downward)}$$

15. Ans. (B)

Sol. It is equivalent to balanced wheat stone bridge

$$R = \frac{L}{KA}$$



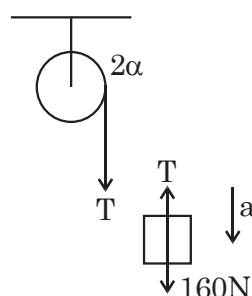
$$R_x = \frac{L}{KA} = R_0$$

$$R_y = \frac{L}{2KA} = \frac{R_0}{2}$$

$$R_a = \frac{R_0}{2} + \frac{2R_0 R_0}{3R_0}$$

$$= R_0 \left[\frac{1}{2} + \frac{2}{3} \right] = \frac{7R_0}{6} = \frac{7}{6} \cdot \frac{L}{KA}$$

16. Ans. (B)



Sol.

$$160 - T = 16a \quad \dots\dots(1)$$

$$T(0.2) = 0.5(\alpha) \quad \{a = \alpha R\}$$

$$T(0.2) = 0.5 \left(\frac{a}{0.2} \right)$$

$$T = 2.5 \left[\frac{a}{0.2} \right] = 12.5a \quad \dots\dots(2)$$

From eqn. (1) and (2)

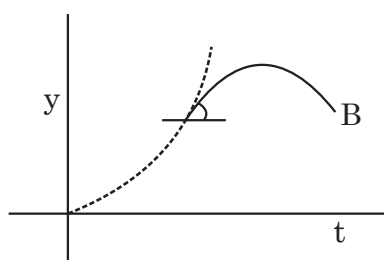
$$160 = T + 16a = 28.5a$$

$$a = \frac{160}{28.5} \text{ m/s}^2$$

17. Ans. (B)

Sol. as bolt looses from ceiling. It still have vel. equal to vel. of lift at that moment. But after this, Vel. of bolt decreases bcoz of gravity

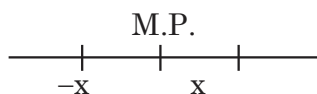
and after some time it moves downward.



18. **Ans. (A)**

Sol. $v = \omega \sqrt{A^2 - X^2}$

Speed is same at same separation from M.P.,



So KE at x = KE at $-x$

So PE at x = PE at $-x$

KE = 5J

u = 3J

19. **Ans. (B)**

Sol. Here $T_3 > T_2 > T_1$

$\Delta U = nC_v \Delta T$

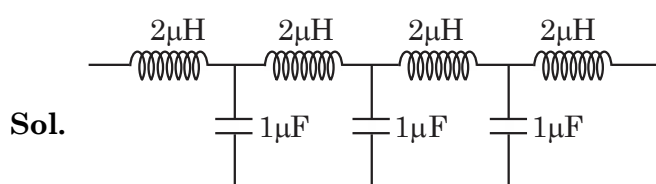
Here ΔT is maximum in process B.

20. **Ans. (A)**

Sol. Maximum frequency is required for material 3, so $\phi_3 > \phi_2 > \phi_1$.

SECTION-II

1. **Ans. 1.41**



Sol.

$$Z = X_L + \frac{ZX_C}{Z + X_C}$$

$$Z(Z + X_C) = X_L(Z + X_C) + ZX_C$$

$$Z^2 + ZX_C = ZX_L + X_L X_C + ZX_C$$

$$Z^2 - ZX_L - X_L X_C = 0$$

Solving,

$$Z = \omega LC \pm \sqrt{(\omega LC)^2 - 4LC}$$

For pure inductive nature, no X_C should exist

$$\text{So, } \omega^2 L^2 C^2 - 4LC = 0 \text{ or } \omega = \frac{2}{\sqrt{LC}}$$

$$\omega = \frac{2}{(2 \times 10^{-16} \times 1 \times 10^{-6})^{1/2}} = \sqrt{2} \times 10^6 \text{ rad/s}$$

2. **Ans. 10.00**

Sol. $v_A = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{4.8}{1.2 \times 10^{-2}}} = 20 \text{ m/s}$

$$v_B = \sqrt{\frac{7.5}{1.2 \times 10^{-2}}} = \sqrt{\frac{2500}{4}} = \frac{50}{2} = 25 \text{ m/s}$$

Separation = $v_{\text{rev}} (\Delta t)$

$$20(20) \times 10^{-3} = 5(\Delta t)$$

$$(\Delta t) = 80 \times 10^{-3} = 8 \times 10^{-2}$$

$$t_{\text{final}} = 20 \text{ ms} + 80 \text{ ms}$$

$$= 100 \text{ ms}$$

$$= 10 \times 10^{-2} \text{ sec}$$

3. **Ans. 13.33**

4. **Ans. 4.50**

Sol. $W_{\text{net}} = K_f - K_i$

$$W_B + W_{\text{mg}} + W_{\text{fr}} = 0 - \frac{1}{2}(0.05)(20)^2$$

$$F_B = \rho_L v_d g = 2\rho_s v_d g = 2mg$$

$$-2mg(2) + mg(2) + w = -10$$

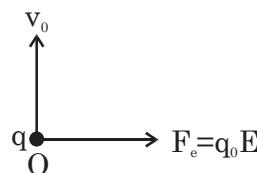
$$w = 2mg - 10 = \frac{2(50)}{1000} \times 10 - 10 = -9 \text{ J}$$

$$-W = -9 \text{ J}$$

$$\frac{W}{2} = \frac{9}{2} = 4.50$$

5. **Ans. 1.73**

Sol. $\lambda = \frac{h}{p}$



$$p_i = mv_0$$

To half the wavelength

p has to be doubled

$$v_f = 2v_0 = \sqrt{v_x^2 + v_y^2}$$

$$4v_0^2 = \left[\frac{q_0 E}{m} t \right]^2 + v_0^2$$

$$3v_0^2 = \frac{q_0^2 E^2}{m^2} \cdot t^2$$

$$t = \frac{mv_0}{q_0 E_0} \sqrt{3}$$

$$n = \sqrt{3} = 1.73$$

6. Ans. 2.50

Sol. $\vec{j} = (5-2)\hat{i} + (4-0)\hat{j} + (2-2)\hat{k}$

$$\vec{j} = 3\hat{i} + 4\hat{j}$$

$$\vec{F} = 6\hat{i} + 8\hat{j}$$

$$w = 18 + 32 = 50J$$

7. Ans. 1.00

Sol. $\mu = \sqrt{2}$

for min. deviation

$$i = e$$

$$r_1 = r_2 = A/2$$

$$\text{So } \delta = i + e - A$$

$$\delta_{\min} = 2i - A$$

$$\alpha = 2i - \alpha$$

$$\boxed{i = \alpha}$$

Now apply Snell's law

$$(1) \sin i = \sqrt{2} \sin(r)$$

$$1 \sin(\alpha) = \sqrt{2} \sin\left(\frac{\alpha}{2}\right)$$

$$\sin \alpha = \sqrt{2} \sin\left(\frac{\alpha}{2}\right)$$

$$2 \sin\left(\frac{\alpha}{2}\right) \cdot \cos\left(\frac{\alpha}{2}\right) = \sqrt{2} \sin\left(\frac{\alpha}{2}\right)$$

$$\cos\left(\frac{\alpha}{2}\right) = \frac{1}{\sqrt{2}}$$

$$\frac{\alpha}{2} = 45^\circ$$

$$\alpha = 90^\circ$$

$$\text{So } \sin(\alpha) = \sin(90^\circ) = 1$$

8. Ans. 1.50

Sol. $F^a A^b T^c = M^0 L^1 T^0$

$$[M^1 L^1 T^{-2}]^a [L^1 T^{-2}]^b [T^1]^c = L^1$$

$$M^a L^{a+b} T^{-2a-2b+c} = M^0 L^1 T^0$$

$$a = 0$$

$$a + b = 1 \Rightarrow b = 1$$

$$-2a - 2b + c = 0$$

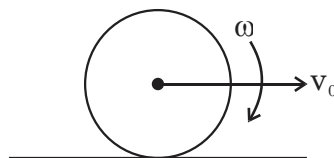
$$0 - 2 + c = 0$$

$$c = 2$$

$$\frac{a+b+c}{2} = \frac{0+1+2}{2} = \frac{3}{2}$$

9. Ans. 1.50

Sol.



$$\omega = \frac{v}{R}$$

$$L = mv_c R \otimes + I_{cm} \omega \otimes$$

$$= m(\omega R)(R) + \frac{mR^2}{2} \cdot \omega$$

$$\boxed{L = \frac{3}{2} m \omega R^2}$$

10. Ans. 4.09 to 4.11

Sol. Given $\frac{\lambda}{2} = 20\text{cm}$

$$\lambda = 40\text{cm} = 0.4\text{m}$$

$$v = f \lambda = 1024(0.4)$$

$$v = 409.6 \text{ m/s}$$

$$\text{So } \frac{v}{100} = 4.096 \approx 4.10$$

SECTION-I

1. Ans. (B)

Sol. $m = \frac{\text{No. moles of solute}}{\text{mass of solvent}} \times 1000$

$$m = \frac{w_2}{M_2} \times \frac{1000}{\frac{w_1}{M_1} \times M_1}$$

$$m = \frac{n_2 \times 1000}{n_1 \times M_1}$$

$$\chi_{\text{solvent}} = 0.7, \chi_{\text{solute}} = 0.3$$

(n_1) (n_2)

$$m = \frac{0.3 \times 1000}{0.7 \times 18} = 23.80 \text{ mol kg}^{-1}$$

2. Ans. (B)

3. Ans. (B)

4. Ans. (C)

5. Ans. (C)

6. Ans. (A)

7. Ans. (C)

Sol. De-Broglie wavelength (λ) = $\frac{h}{p}$

$$\text{K.E. of } e^- = eV$$

$$\text{K.E.} = P^2/2m$$

$$eV = P^2/2m$$

$$P = \sqrt{2meV}$$

$$\frac{h}{\lambda} = P = \sqrt{2meV}$$

8. Ans. (D)

9. Ans. (A)

10. Ans. (C)

Sol. In Frenkel defect cation leave their original position and occupy interstitial site.

11. Ans. (D)

12. Ans. (B)

Sol. β -keto acid undergo decarboxylation in very mild condition.

13. Ans. (A)

14. Ans. (B)

15. Ans. (A)

16. Ans. (A)

Sol. $\text{pH} = C\alpha = 10^{-4}$

$$\alpha = \frac{10^{-4}}{0.0015} = \frac{1}{15}$$

$$K = G \left(\frac{\ell}{A} \right)$$

$$= 5 \times 10^{-7} \times \frac{120}{1} = 6 \times 10^{-5} \text{ Scm}^{-1}$$

$$\Lambda_m = \frac{K \times 1000}{C} = \frac{6 \times 10^{-5} \times 1000}{0.0015}$$

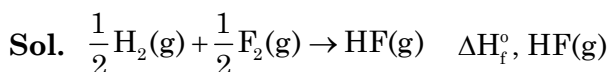
$$\Lambda_m^0 = \frac{\Lambda_m}{\alpha} = \frac{6 \times 10^{-5} \times 1000}{0.0015} \times \frac{0.0015}{10^{-4}}$$

$$= 600 \text{ S cm}^2 \text{ mol}^{-1}.$$

17. Ans. (A)

18. Ans. (A)

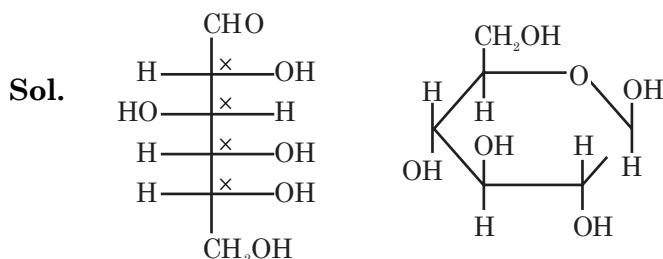
19. Ans. (B)



20. Ans. (A)

SECTION-II

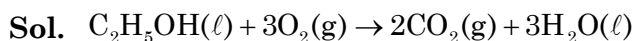
1. Ans. (9.00)



No. of stereo centre
= 4 = X

No. of stereo centre
(Y) = 5

2. Ans. (326.40)



$$\Delta n_g = 2 - (3) = -1$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$-327000 = \Delta U + (-1)(2) \times (300)$$

$$\Delta U = -326.40 \text{ cal.}$$

3. Ans. (3.00)

4. Ans. (4.00)

Sol. Gabriel phthalamide is used to synthesise 1° aliphatic/acyclic amine.

5. Ans. (7.57 or 7.58)

Sol. $\text{Pb} + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4 + 2\text{e}^-$

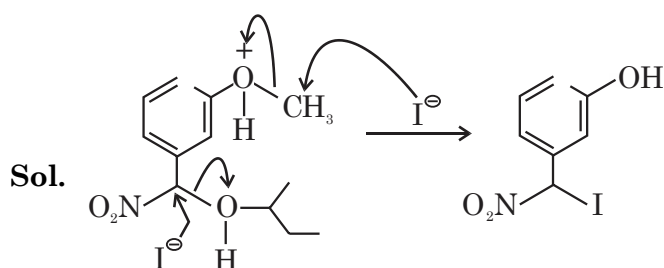
If $\equiv$ 1g equiv. of PbSO_4

$$= \frac{1}{2} \text{mol of PbSO}_4 \Rightarrow \frac{303}{2} \text{g PbSO}_4$$

$$0.05 \text{F} = \frac{303}{2} \times 0.05 \text{g of PbSO}_4 = 7.57 \text{g}$$

6. Ans. (7.00)

7. Ans. (5.00)



8. Ans. (9.00)

Sol. $\text{H}_2\text{SO}_4 + 2\text{NH}_4\text{OH} \rightarrow (\text{NH}_4)_2\text{SO}_4 + 2\text{H}_2\text{O}$

$$t = 0 \quad 20 \times 0.1 = 2 \quad 30 \times 0.2 = 6 \quad - \quad -$$

$$t = t' \quad 0 \quad 2 \quad 2$$

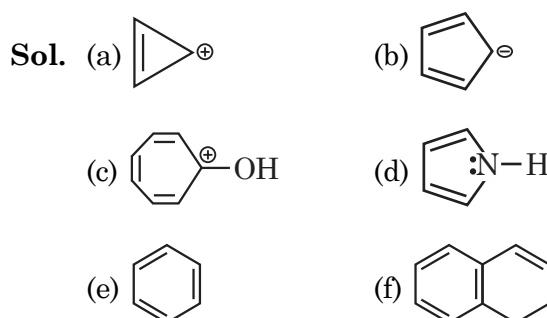
Resulting solution is buffer.

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{NH}_4^+]_0}{[\text{NH}_4\text{OH}]_0} = 4.7 + \log \frac{4}{2} = 5.0$$

$$\text{pH} = 14 - \text{pOH} = 9.0$$

9. Ans. (4.00)

10. Ans. (6.00)



PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (C)

Sol.

p	q	$\sim p$	$\sim q$	$\sim p \leftrightarrow q$	$q \vee p$	$p \leftrightarrow \sim q$	$\sim(p \leftrightarrow \sim q)$
T	T	F	F	F	T	F	T
T	F	F	T	T	T	T	F
F	T	T	F	T	T	T	F
F	F	T	T	F	F	F	T

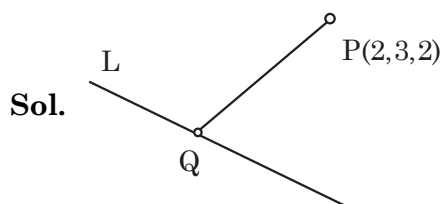
2. Ans. C

Sol. for $m = 1$ lines are concurrent.

for $m = -1/2$ and $m = 2$, two of the lines are parallel

$\therefore$ one of the possible value of m can be -2 .

3. Ans. B



Direction vector of line L (intersection of plane)

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 2 \\ 1 & 2 & -1 \end{vmatrix} = \hat{i}(-5) - \hat{j}(-1) + \hat{k}(-3)$$

one of fixed point $(-2, 2, 0)$

$$\text{equation of line L: } \frac{x+2}{5} = \frac{y-2}{-1} = \frac{z}{3}$$

Let point Q: $(-2+5\lambda, 2-\lambda, 3\lambda)$

$$\overrightarrow{PQ} = (5\lambda - 4)\hat{i} + (-\lambda - 1)\hat{j} + (3\lambda - 2)\hat{k}$$

$\overrightarrow{PQ}$ is parallel to $\vec{r}$.

$$\frac{5\lambda - 4}{1} = \frac{-\lambda - 1}{-2} = \frac{3\lambda - 2}{1}$$

$$\lambda = 1$$

required distance

$$= |\overrightarrow{PQ}| = \sqrt{1^2 + 2^2 + 1^2} = \sqrt{6}$$

4. **Ans. C**

Sol. $x^2 - xy + y = 0$

$$2x - x \frac{dy}{dx} - y + \frac{dy}{dx} = 0$$

$$2x - y = (x-1) \frac{dy}{dx}$$

$$\Rightarrow 2x - y = (x-1) \left(\frac{-dx}{dy} \right)$$

$$\Rightarrow \frac{dy}{dx} = - \left(\frac{x-1}{2x-y} \right)$$

$$\Rightarrow y = g(x) \Rightarrow g'(0) = -\frac{1}{2}$$

$\therefore$ equation of tangent

$$y - 2 = -\frac{1}{2}(x - 0)$$

$$\Rightarrow x - \text{intercept} = 4$$

5. **Ans. C**

Sol.
$$\begin{vmatrix} 2 & -1 & 1 \\ 1 & \sin \theta & 2 \cos \theta \\ 2 & -1 & (\sin \theta - \cos \theta) \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 2 & -1 & 1 \\ 1 & \sin \theta & 2 \cos \theta \\ 0 & 0 & (\sin \theta - \cos \theta - 1) \end{vmatrix} = 0$$

$$\Rightarrow (\sin \theta - \cos \theta - 1)(2 \sin \theta + 1) = 0$$

$$\Rightarrow \sin \theta = -\frac{1}{2}, \sin \theta - \cos \theta = 1$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}, \theta = \frac{\pi}{2}, \pi$$

$$\therefore 3\pi + \frac{3\pi}{2} = k\pi \Rightarrow k = \frac{9}{2}$$

6. **Ans. (C)**

Hence $f(x)$ is continuous at $x = \frac{\pi}{2}$

$$\Rightarrow k = \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{(\pi - 2x)^2} \cdot \frac{\log \sin x}{\log(1 + \pi^2 - 4\pi x + 4x^2)}$$

Put $\alpha = \left(\frac{\pi}{2} - x \right)$

$$k = \lim_{\alpha \rightarrow 0} \frac{1 - \cos \alpha}{(2\alpha)^2} \cdot \frac{\log \cos \alpha}{\log(1 + 4\alpha^2)}$$

$$= \lim_{\alpha \rightarrow 0} (1 - \cos \alpha) \frac{\log(1 + \cos \alpha - 1)}{(\cos \alpha - 1)} \frac{4\alpha^2}{\log(1 + 4\alpha^2)} \frac{(\cos \alpha - 1)}{(2\alpha)^2 4\alpha^2}$$

$$= \lim_{\alpha \rightarrow 0} \frac{1 - \cos \alpha}{\left(\frac{\alpha^2}{4} \right)} \frac{(\cos \alpha - 1)}{\frac{\alpha^2}{4}} \frac{\frac{\alpha^2}{4} \cdot \frac{\alpha^2}{4}}{4\alpha^2 \cdot 4\alpha^2}$$

$$k = \frac{2 \cdot (-2)}{4 \cdot 4 \cdot 4 \cdot 4} = -\frac{1}{64}$$

7. **Ans. D**

Sol. Let point on L, $(-2 + \lambda, 2 - 2\lambda, -1 + 3\lambda)$

Point of intersection,

$$(-2 + \lambda) + (2 - 2\lambda) + (-1 + 3\lambda) = 11$$

$$\Rightarrow \lambda = 6$$

Intersecting point (P): $(4, -10, 17)$

Image point:

$$\frac{x-4}{-2} = \frac{y+10}{1} = \frac{z-17}{2} = \frac{-2[-8-10+34-6]}{9}$$

$$= \frac{-20}{9}$$

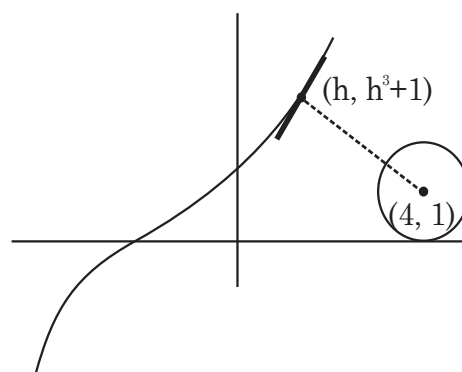
$$x = \frac{40}{9} + 4 = \frac{76}{9}$$

$$y = \frac{-20}{9} - 10 = \frac{-110}{9}$$

$$z = \frac{-40}{9} + 17 = \frac{113}{9}$$

8. **Ans. C**

Sol. $y = x^3 + 1$



$$y' = 3x^2$$

$$y' \big|_{(h, h^3+1)} = 3h^2$$

$$\therefore -\frac{1}{3h^2} = \frac{h^3 + 1 - 1}{h - 4}$$

$$-(h - 4) = 3h^5$$

$$3h^5 + h = 4$$

9. Ans. (A)

$$I = \int \left(\frac{1 - \tan^2 x}{1 + \tan^2 x} \right) \frac{1}{\sin 2x} dx = \int \frac{\cos 2x}{\sin 2x} dx$$

$$= \int \cot 2x dx = \frac{\ln |\sin 2x|}{2} + C$$

10. Ans. A

Sol. $S = {}^{50}C_6 - {}^5C_1 {}^{40}C_6 + \dots + {}^5C_4 {}^{10}C_6$

$$= \text{Coefficient of } x^6 \text{ in } {}^5C_0 (1+x)^{50}$$

$$- {}^5C_1 (1+x)^{40} + \dots + {}^5C_4 (1+x)^{10} - {}^5C_5 (1+x)^0$$

$$= \text{Coefficient of } x^6 \text{ in } ((1+x)^{10} - 1)^5$$

$$= \text{Coefficient of } x^6 \text{ in } x^5 ({}^{10}C_1 + {}^{10}C_2 x + \dots + {}^{10}C_{10} x^9)^5$$

$$= {}^5C_4 ({}^{10}C_1)^4 ({}^{10}C_2)$$

$$= 5 \times 10^4 \times 5 \times 9$$

$$= 2250000$$

11. Ans. B

Sol. $f(x) = x^3 + x$

$$f'(x) = 3x^2 + 1 \Rightarrow f(x) \text{ is } \uparrow$$

$$f(f(x)) > f(4-2x) \Rightarrow f(x) > 4-2x$$

$$\Rightarrow x^3 + x > 4-2x \Rightarrow x^3 + 3x - 4 > 0$$

$$\Rightarrow x^3 - x + 4x - 4 > 0$$

$$\Rightarrow x(x^2 - 1) + 4(x-1) > 0$$

$$\Rightarrow (x-1)(x^2 + x + 4) > 0 \Rightarrow x > 1$$

Least integral value of x is 2.

12. Ans. (B)

$f \circ g = g \circ f$

$$(ax+b)^2 + (ax+b) - 2 = a(x^2+x-2) + b$$

$$(a^2-a)x^2 + (2ab)x + b^2 - 2 + 2a = 0$$

$\therefore$ equation is an identity

$$\Rightarrow a = 1 \text{ \& } b = 0 \text{ or } a = 0 \text{ \& } b = \pm\sqrt{2}$$

13. Ans. B

Sol. Probability of getting 2 heads in the first 5 trials is

$${}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 = \frac{10}{32} = \frac{5}{16}$$

Therefore, the probability that third head appears on the sixth trial is $\frac{5}{16} \times \frac{1}{2} = \frac{5}{32}$

14. Ans. B

Sol. Number of subsets = ${}^{10}C_3 \times 3 + ({}^{10}C_1)^3$

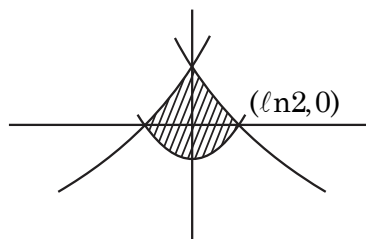
$$= 360 + 1000 = 1360$$

15. Ans. A

Sol. $y = e^{-|x|} - \frac{1}{2}$

$$y = -\sqrt{(\ln 2)^2 - x^2}$$

$$x^2 + y^2 = (\ln 2)^2$$



Reqd. area

$$= 2 \int_0^{\ln 2} \left(e^{-x} - \frac{1}{2} \right) dx + \frac{\pi (\ln 2)^2}{2}$$

$$= 2 \left(-e^{-x} - \frac{1}{2} x \right)_0^{\ln 2} + \frac{\pi (\ln 2)^2}{2}$$

$$= 2 \left(-e^{-\ln 2} - \frac{1}{2} \ln 2 + 1 \right) + \frac{\pi (\ln 2)^2}{2}$$

$$= 1 - \ln 2 + \frac{\pi (\ln 2)^2}{2}$$

$\therefore p = 1$

16. Ans. D

Sol. $17a^2 + 89a + 1 = 0 \dots (1)$

and $b^2 - 89b + 17 = 0 = 1 - \frac{89}{b} + \frac{17}{b^2} = 0$

$$\Rightarrow 17 \left(\frac{-1}{b} \right)^2 + 89 \left(\frac{-1}{b} \right) + 1 = 0 \dots (2)$$

from (1) and (2)

$$a + \left(\frac{-1}{b} \right) = \frac{-89}{17}$$

and $\frac{-a}{b} = \frac{1}{17} \Rightarrow \frac{a}{b} = \frac{-1}{17}$

$$\Rightarrow \left| \frac{ab - 4a - 1}{b} \right| = \left| a - \frac{1}{b} - 4 \frac{a}{b} \right|$$

$$= \left| -\frac{89}{17} + \frac{4}{17} \right| = 5$$

17. Ans. D

Sol. $\triangle ABC$ is equilateral

$$r = \frac{R}{2} = \frac{3}{2}$$

$$a = 2R \sin A = 2.3 \frac{\sqrt{3}}{2} = 3\sqrt{3}$$

$$\therefore (abc)^{1/r} = (a^3)^{1/r} = (3\sqrt{3})^{3/3} = 27$$

18. Ans. (C)

Sol. Using L'Hospital Rule

$$\lim_{x \rightarrow 0} \frac{\ln(1 + \sin^6 x) 2 \sin x \cdot \cos x}{(e^x - 1 - x)(\tan x - x)(1 - \cos x)}$$

$$\lim_{x \rightarrow 0} \frac{\frac{\ln(1 + \sin^6 x)}{\sin^6 x} \times \frac{2 \sin^7 x}{x^7} \cdot \cos x \cdot x^7}{\left(\frac{x^2}{2!} + \frac{x^3}{3!} + \dots\right) \left(\frac{x^3}{3} + \frac{2x^5}{15} + \dots\right) \left(\frac{x^2}{2!} - \frac{x^4}{4!} + \dots\right)}$$

$$= \frac{1 \times 2 \times 1}{\frac{1}{2} \times \frac{1}{3} \times \frac{1}{2}} = 24$$

19. Ans. (C)

Sol. $A(\text{adj } A) = |A| I = 5B^2$

$$\therefore B^2 = I \Rightarrow \sec^2 \theta = 1, \sec \theta = \pm 1, \theta = 0, \pi, 2\pi$$

$$(\sin \theta + \cos \theta)^2 = 1, \sin 2\theta = 0$$

$$2\theta = n\pi \Rightarrow \theta = \frac{n\pi}{2}, n \in I, \theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

common values of θ are

$0, \pi, 2\pi$. Hence 3 values

20. Ans. (B)

$$y = mx - 2m - m^3$$

$$\Rightarrow m^3 + m(2 - x) + y = 0$$

$$m_1 + m_2 + m_3 = 0 \quad \dots\dots\dots(1)$$

$$\sum m_1 m_2 = 2 - x \quad \dots\dots\dots(2)$$

$$m_1 m_2 m_3 = -y \quad \dots\dots\dots(3)$$

$$m_1 m_2 = -1$$

$$m_3 = y$$

$$\text{From (1) } m_1 + m_2 = -y$$

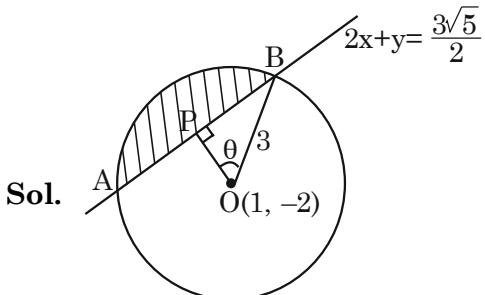
$$\text{From (2) } -1 + y(m_1 + m_2) = 2 - x$$

$$-1 - y^2 = 2 - x$$

$$y^2 = x - 3$$

SECTION-II

1. Ans. 13.00



Sol.

$$OP = \left| \frac{3\sqrt{5}/2}{\sqrt{5}} \right| = \frac{3}{2}$$

$$\cos \theta = \left| \frac{3/2}{3} \right| = \frac{1}{2};$$

$$\therefore \theta = 60^\circ$$

$$\begin{aligned} \text{Required area} &= \frac{\pi(3)^2 \times 120^\circ}{360^\circ} - \frac{1}{2} \times 3^2 \sin 2\theta \\ &= \frac{9\pi}{3} - \frac{9}{2} \cdot \frac{\sqrt{3}}{2} = 9 \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right] \end{aligned}$$

$$\alpha = 9; \beta = 4$$

$$\therefore \alpha + \beta = 13$$

2. Ans. 4.00

$$\text{Sol. } I = \int_0^{\pi/4} (2e^x + (e^x + e^{-x}) \sec^2 x) e^{\tan x + x} dx$$

$$= \int_0^{\pi/4} ((e^x - e^{-x}) + (e^x + e^{-x}) + (e^x + e^{-x}) \sec^2 x) e^{\tan x + x} dx$$

$$= \int e^x \left(\frac{(e^x + e^{-x})e^{\tan x}}{f(x)} + \frac{e^{\tan x}(e^x - e^{-x}) + (e^x + e^{-x})e^{\tan x} \cdot \sec^2 x}{f'(x)} \right) dx$$

$$= (e^x (e^x + e^{-x}) e^{\tan x})_0^{\pi/4} = e^{\pi/4} (e^{\pi/4} + e^{-\pi/4}) e^1 - 2$$

$$= e^1 (e^{\pi/2} + 1) - 2 \equiv e^a (e^{\pi/2} + b) - c$$

$$\therefore a + b + c = 1 + 1 + 2 = 4$$

3. Ans. 8.00

Sol. Given set is

$$\{(a, b) : 2a^2 + 3b^2 = 35, a, b \in \mathbb{Z}\}$$

We can set that

$$2(\pm 2)^2 + 3(\pm 3)^2 = 35$$

$$2(\pm 4)^2 + 3(\pm 1)^2 = 35$$

$(2, 3), (2, -3), (-2, -3), (-2, 3), (4, 1), (4, -1), (-4, -1), (-4, 1)$ are 8 order pair.

4. Ans. 0.60

From skew symmetric matrix

$$a\alpha^2 + b\alpha + c = 0, a\beta^2 + b\beta + c = 0$$

$$a\gamma^2 + b\gamma + c = 0$$

$$\Rightarrow ax^2 + bx + c = 0 \text{ has three roots}$$

$$a = b = c = 0$$

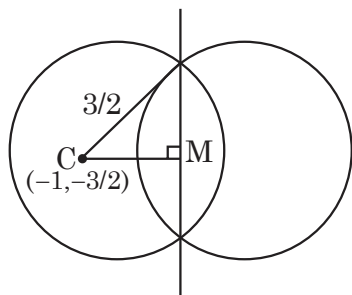
$$\lambda = \begin{vmatrix} 1 & 1 & 2 \\ 3 & 4 & 1 \\ 9 & 0 & 3 \end{vmatrix} = 12 + 2(-36) = 60$$

5. Ans. 2.00

Sol. Equation of common chord $2x + 1 = 0$

$$C\left(-1, \frac{-3}{2}\right), r^2 = \frac{9}{4}, CM^2 = \left|\frac{2(-1)+1}{2}\right|^2$$

$$AB = 2\sqrt{r^2 - CM^2} = 2\sqrt{\frac{9}{4} - \frac{1}{4}} = 2\sqrt{2}, [2\sqrt{2}] = 2$$



6. Ans. 6.00

Sol. $y(x^2 - 3|x| + 9) = 18|x|$

$$y = \frac{18|x|}{x^2 - 3|x| + 9} = \frac{18}{|x| + \frac{9}{|x|} - 3}$$

$$y \leq \frac{18}{6-3}, y \leq 6 \Rightarrow y_{\max} = 6$$

7. Ans. 2.00

Sol. $L = \lim_{x \rightarrow 0} \frac{(\sin x - x)^2 + 1 - \cos x^3}{x^5 \sin x}$

$$= \lim_{x \rightarrow 0} \frac{(\sin x - x)^2}{x^5 \sin x} + \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x^3}{2}}{x^5 \sin x}$$

$$= \lim_{x \rightarrow 0} \frac{\left(\left(x - \frac{x^3}{3} + \dots\right) - x\right)^2}{x^5 \sin x} + \frac{1}{2}$$

$$= \frac{1}{36} + \frac{18}{36} = \frac{19}{36}$$

8. Ans. 3.00

Sol. $\vec{r} \times \vec{a} = \vec{c} \times \vec{a}$

$$\vec{b} \times (\vec{r} \times \vec{a}) = \vec{b} \times (\vec{c} \times \vec{a})$$

$$(\vec{a} \cdot \vec{b})\vec{r} - (\vec{b} \cdot \vec{r})\vec{a} = (\vec{a} \cdot \vec{b})\vec{c} - (\vec{b} \cdot \vec{c})\vec{a}$$

$$[-4]\vec{r} = -4\vec{c} - 4\vec{a}$$

$$\vec{r} = \vec{c} + \vec{a} = \hat{i} + 2\hat{j} - \hat{k}$$

$$[\vec{r} \ \vec{a} \ \vec{b}] = \begin{vmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 1 & -1 & -1 \end{vmatrix} = 1 - 2(-1) - 1(0) = 3$$

Aliter : $(\vec{r} \times \vec{a}) \cdot \vec{b} = (\vec{c} \times \vec{a}) \cdot \vec{b} = [\vec{c} \ \vec{a} \ \vec{b}]$

9. Ans. 2.50

Sol. Let A and B be two independent events

$$P(A) = \frac{2}{7}, P(B) = \frac{6}{11}$$

P(occurrence of atleast one of the events)

$$= 1 - P(\text{none})$$

$$= 1 - P(\bar{A})P(\bar{B})$$

$$= 1 - \frac{5}{7} \cdot \frac{5}{11} = \frac{52}{77}. \text{ Hence } \left(\frac{q-p}{10}\right) = 2.50$$

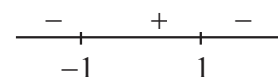
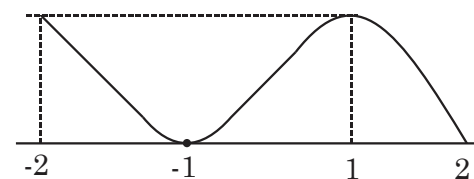
10. Ans. 4.00

$$\text{Sol. } AB = \begin{bmatrix} -x^3 + 4 - 6 & - & - \\ - & 7 + 2x - 12 & - \\ - & - & 9 + x \end{bmatrix}$$

$$T_r(AB) = -x^3 - 2 + 2x - 5 + 9 + x$$

$$f(x) = -x^3 + 3x + 2$$

$$f'(x) = -3x^2 + 3 = -3(x^2 - 1)$$



$$f(-1) = 0$$

$$f(1) = 4$$

$$f(2) = -8 + 6 + 2 = 0$$

$$f(-2) = 8 - 6 + 2 = 4$$

$$\therefore \text{ for } x \in [-2, 2], f(x)_{\max} = 4$$